Crontab Assignemnt

Task 1: Three real-life uses of crontab

You can write these examples directly in your assignment.

1. Automatic backup: Schedule daily backups of important files to prevent data loss if files are accidentally deleted or the system fails.
2. Log file cleanup: Schedule a script to delete old log files regularly to save disk space.
3. System monitoring: Run a script every few minutes to check CPU, memory, or disk usage and record the results for troubleshooting.

Task 2: Run update_playlist.sh every day at 2:00 AM

Step 1: Open your crontab

Run this as your normal Linux user:

crontab -e

If it asks you to select an editor, choose nano if available.

Step 2: Add this entry

0 2 * * * /home/diya/update_playlist.sh

Replace diya with your actual username if different.

Meaning of each part

Field	Value	Meaning
Minute	0	At minute zero
Hour	2	At 2 AM
Day of month	*	Every day of the month
Month	*	Every month
Day of week	*	Every day of the week
Command	Script path	Runs the script

Save and exit. In Nano, press Ctrl+O, Enter, then Ctrl+X.

Note: The script must exist and be executable for this scheduled job to work.

Task 3: Create backup_instagram_photos.sh

We'll create a script that copies your Instagram photo files into a new backup folder named with today's date, for example, InstagramPhotos_2026-10-20.

Step 1: Create the required directories

Run:

mkdir -p ~/InstagramPhotos

mkdir -p ~/Backups

These commands create the source and backup directories if they do not already exist.

Step 2: Create the shell script

Run:

nano ~/backup_instagram_photos.sh

Copy and paste this script:

```
#!/bin/bash

SOURCE="$HOME/InstagramPhotos"
BACKUP="$HOME/Backups"
DATE=$(date +%F)
DEST="$BACKUP/InstagramPhotos_$DATE"

echo "[$(date)] Starting Instagram photos backup"

if [ ! -d "$SOURCE" ]; then
    echo "ERROR: Source directory does not exist: $SOURCE"
    exit 1
fi

mkdir -p "$DEST" || {
    echo "ERROR: Could not create backup directory: $DEST"
    exit 1
}

if cp -a "$SOURCE/." "$DEST/"; then
    echo "SUCCESS: Photos copied to $DEST"
else
    echo "ERROR: Backup copy failed"
    exit 1
fi
```

echo "[$(date)] Backup completed"

Save the file using Ctrl+O, press Enter, then Ctrl+X.

Step 3: Give the script execution permission

chmod +x ~/backup_instagram_photos.sh

Verify:

ls -l ~/backup_instagram_photos.sh

The permissions should include x, meaning the script is executable.

Step 4: Run the script manually for testing

~/backup_instagram_photos.sh

Expected output will resemble:

Starting Instagram photos backup

SUCCESS: Photos copied to /home/diya/Backups/InstagramPhotos_2026-10-20

Backup completed

Check the backup directory:

ls -l ~/Backups/

Task 3 is complete when the dated backup folder is created successfully. If the source directory is empty, the backup can still succeed, but no photos will be copied.

Task 4: Schedule the backup every Sunday at 11:30 PM

Step 1: Open your crontab

crontab -e

Add this entry on a new line:

30 23 * * 0 /home/diya/backup_instagram_photos.sh >> /home/diya/backup_log.txt 2>&1

Replace diya with your actual username if needed.

Step 2: Understand the schedule and logging

Part	Meaning
30	Minute 30
23	Hour 23 (11 PM)
*	Every day of the month
*	Every month
0	Sunday

Script path	Runs the backup script
>>	Appends output to the log file
2>&1	Sends error output to the same log file

This job runs every Sunday at 11:30 PM according to the Linux server's local time.

Save and exit the editor.

Step 3: Verify the scheduled job

crontab -l

You should see the new cron entry.

On RHEL 9, ensure the cron service is running:

sudo systemctl enable --now crond

Verify its status:

systemctl status crond

Look for Active: active (running).

Step 4: Test the log immediately

You don't need to wait until Sunday to test the script and logging. Run:

~/backup_instagram_photos.sh >> ~/backup_log.txt 2>&1

Check the log:

cat ~/backup_log.txt

Task 5: Show the last five lines of the log

Run:

tail -n 5 ~/backup_log.txt

Example output:

[Tue Oct 20 19:30:00 IST 2026] Starting Instagram photos backup

SUCCESS: Photos copied to /home/diya/Backups/InstagramPhotos_2026-10-20

[Tue Oct 20 19:30:00 IST 2026] Backup completed

The example contains only three log lines, so tail -n 5 will show all available lines. Your actual output may differ.

Important: Running the script manually proves that the backup and logging work, but it does not prove that cron has executed the Sunday schedule. To verify the scheduled execution afterward, check the log again after Sunday at 11:30 PM.

On RHEL, you can also inspect cron activity with:

sudo journalctl -u crond